
CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/22**

Paper 2 (Extended)

May/June 2017

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2017 series for most Cambridge IGCSE[®], Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

© IGCSE is a registered trademark.

This document consists of **4** printed pages.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

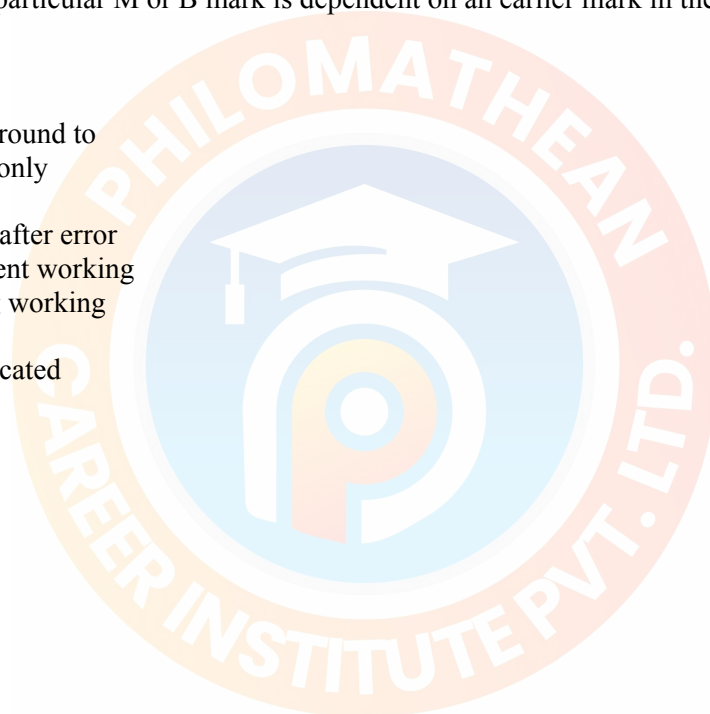
Types of mark

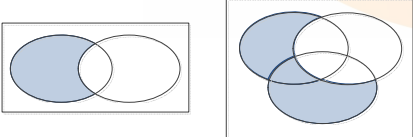
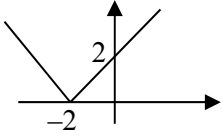
- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘**dep**’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied



Question	Answer	Marks	Partial Marks
1(a)	5.310	1	
1(b)	[0].00365	1	
2(a)	$ \begin{array}{c c} 1 & 7 \\ 2 & 6\ 6\ 8 \\ 3 & 0\ 3\ 4\ 7\ 7 \\ 4 & 1\ 5\ 6\ 6\ 8\ 8\ 9 \\ 5 & 2\ 3\ 3\ 4 \end{array} $ Key 5 3 = 53	3	B2 for all elements correct or B1 for all 'correct' but unordered or 1 or 2 missing or wrongly placed or extra element(s) B1 for key
2(b)	43	1	
3	$2x(3x - 1)$ final answer	2	B1 for $2(3x^2 - x)$ or $x(6x - 2)$
4	0 2	2	B1 for each (must be correct order)
5	$5x + 2$ final answer	2	B1 for $ax + 2$ or $5x + b$
6	$27 + 9\pi$ or $9(3 + \pi)$ or $3(9 + 3\pi)$	3	M1 for $\frac{1}{2} \times 9 \times 6$ oe M1 for $\frac{1}{4} \times \pi \times 6^2$ oe
7(a)	$x \leq 4$ or $4 \geq x$	2	M1 for $2 + 6 \geq 5x - 3x$ oe If 0 scored, SC1 for $x = 4$, $x < 4$, $x > 4$, $x \geq 4$
7(b)	Correct FT from(a) on number line	1	FT dep on inequality as answer to (a)
8(a)	BCD	1	Must be that order
8(b)	9	2	M1 for $\frac{x}{6} = \frac{6}{4}$ oe
9(a)		2	B1 for each
9(b)	$A \cap B \cap C'$ oe	1	
10	$6x^2 - 17xy + 12y^2$ final answer	3	B2 for $6x^2 - 8xy - 9xy + 12y^2$ or B1 for 3 terms correct
11		3	B2 for basic shape with cusp on -ve x-axis or B1 for a basic 'v' shape modulus graph
12(a)	35	1	

Question	Answer	Marks	Partial Marks
12(b)	130	1	FT 165 – <i>their</i> (a)
13	$40\sqrt{3}$	3	M1 for $(p+q)(p-q)$ soi A1 for $p+q=10$ or $p-q=4\sqrt{3}$ OR B1 for $37+20\sqrt{3}$ oe B1 for $37-20\sqrt{3}$ OR B2 for $10\sqrt{3}+10\sqrt{3}+10\sqrt{3}+10\sqrt{3}$
14	4	2	M1 for $\log 2^5$ or $\log \frac{a}{8}$ or $3\log 2$ or $\log 2^3$
15	$a=2$ $b=-2$ $c=-12$	3	B2 for 2 correct or M2 for $k(x+2)(x-3)$ and substitution of (4, 12) or M1 for $(x+2)(x-3)$ soi by $a=1$, $b=-1$, $c=-6$ or three correct equations in a , b , c If 0 scored, B1 for 1 correct.